College Clustering — Segmenting the CEEB Colleges

Companion to the *OPE↔CEEB Merge Report* · Date: 2026-08-02 · Input: `ope_ceeb_scorecard_merged_clean_2026-08-02.csv` · Code: `etl/build_college_clustering.py` · **Exploratory first pass** — pattern detection, not a final model.

1. Objective

The merge delivers **what** each CEEB college is (identity + Scorecard metrics). This step asks **which colleges resemble each other** — grouping them by location, academic profile, price, and student funding, and surfacing the natural segments in the file. It is the secondary-goal “cluster similar institutions / detect patterns” applied directly to the merged data.

2. Universe and method

Universe: 2,381 degree-granting colleges. The ~30% of CEEB rows with no live federal record (closed / renamed / non-Title-IV holders — the same tail documented in the merge report) carry no metrics and are excluded, as are non-degree entities. **Geography was filled from IPEDS HD2023** (already in hand): locale/urbanicity and coordinates, which the clean merge did not carry — this lifts the “location” dimension from state-only to ~100% coverage. Features are z-scored, reduced by PCA, and clustered with k-means (complete-case, no imputation). Admission rate, SAT and completion are *thinly covered by design* (open-admission and 2-year schools are exempt from the IPEDS admissions component), so they are reported as segment *overlays*, not clustering inputs.

The dominant structure is a clean two-way split (public 2-year vs. private 4-year). A six-segment cut is reported below because it is the operationally useful view.

3. The six segments

#	n	Segment (plain-English)	Type	Setting / region	Med. UG	Med. net price	Pell	Adm.	SAT	Compl.
0	729	Large public community colleges	public / assoc.	City · South	7,629	$10,400	31%	76%	1159	51%
1	173	High-need small private colleges (access-oriented; likely MSI/HBCU-heavy)	private / bach.	City · South	536	$13,996	67%	71%	982	37%
2	415	Mid-size regional private colleges	private / bach.	City · South	1,097	$22,870	35%	76%	1142	56%
3	465	Small-town / rural public 2-year (most open-access)	public / assoc.	Town · South	1,710	$10,075	31%	82%	1103	42%
4	300	Selective private colleges (the elite tier)	private / bach.	City · Northeast	3,439	$32,106	19%	52%	1357	78%
5	211	Small Midwest liberal-arts privates	private / bach.	Town · Midwest	958	$22,139	32%	72%	1154	55%

Overlays (admission rate, SAT, completion) are averages over the segment members that report them; they describe the segments, they did not form them.

4. What we have for clustering — and the gaps

Strong (>95% of the degree-granting universe): size, control, degree level, tuition, net price, Pell share, and — after the HD2023 join — state, region, urbanicity, coordinates.

Thin or missing (flagged for the client conversation):

- **Selectivity is thin** — admission rate 62%, SAT 42%. This is structural, not an error: open-admission and 2-year schools are exempt from the IPEDS admissions survey. Kept as an overlay.
- `sc_act_mid` **is empty (0% populated)** in this vintage — dropped.
- **Student funding is a single proxy (Pell share).** Worth adding: % with federal loans, average aid amount — to make “student funding” a real dimension rather than one variable.
- **No outcomes beyond completion/retention** (earnings and median debt are empty in recent Scorecard vintages, as the merge report notes).

5. Suggested next steps

- Cluster **within** 2-year and 4-year separately — the sector split dominates, and finer, more actionable sub-segments emerge inside each.
- Add the funding variables above so the “student funding” dimension is more than Pell.
- Confirm the intended grain with the client: CEEB is the *score-recipient* (can be finer than an institution), while Scorecard reports at the institution level — the merge resolves to institution grain.

Backing data: `college_cluster_profiles_fine_2026-08-02.csv` (the table above) and `college_clustering_coverage_2026-08-02.csv` (per-feature coverage). Full per-college assignments and an interactive map are in the project dashboard.